

flightcollapse v0.6.3

How a transcript model is built from aligned long reads — reference-anchored collapse for oligo-dT single-cell Iso-Seq

reference-anchored — keeps a GENCODE identity merged into its parent mono-exon track novel model rejected new or returned in 0.6.x panels 11–13 are new

In every icon the grey track is GENCODE; the coloured track is what flightcollapse emits. Faint dashed boxes are reference exons the model does not contain. Support is counted in MOLECULES — UMI-collapsed per cell, not in reads. 0.6.x changes NOTHING in panels 1–9. It adds a final consolidation of the finished catalogue (panels 11–13), which merges same-chain models at the 3' end and folds 5' fragments into their containers. - no -posthoc reproduces 0.5.1 exactly. Numbers on this sheet are measured on the nine 0.5.1 runs: BD144_kin_HMND and BD166b/BD174a/BD176c/BD178b/BD67/BD70.

FIRST PASS builds the catalogue · unchanged since 0.4.6 · reproduce it exactly with - no -second-pass

1 Read admission

One definition of an admitted read, shared by the scanning pass and the output pass, so the two can never disagree.

admitted rejected
coverage ≥ 0.90 · identity ≥ 0.95 · 5' soft clip ≤ 200 nt

On an untrimmed BAM the poly(A) is soft-clipped, so a 0.99 coverage gate keeps one read in six and enriches for mono-exonic ones. The pipeline therefore trims, deduplicates, then re-materialises the founders untrimmed — the tail has to be there for panel 9.

2 Junction census

Every junction in every admitted read is counted once, with its motif, flanking repeat, share of the traffic at each of its own splice sites, and distance to the nearest annotated site.

GENCODE
read
snapped
annotation-first: a read site within 5 bp moves onto the annotated site

internal deletions are promoted first

A short intron spelled as a CIGAR D rather than an N leaves the intron chain untouched, so a real splicing event can hide inside an FSM. Gaps of 20–5,000 bp seen in ≥ 3 reads are promoted to junctions before the census, then curated like any other.

does the read have any intron?

YES → spliced track, panels 3–5 · NO → mono-exon track, panel 6

The empty chain is not a suffix of anything. That single rule is the mono-exon fix; unspliced reads skip panels 3–5 entirely.

3 Junction curation

Junctions are curated BEFORE chains are built. A chain with 40 reads containing one spurious junction survives a chain threshold; it does not survive a junction threshold.

An ANNOTATED junction is never rejected — dropping one because it is rare in a single sample would fabricate novel chains out of well-known transcripts.

a junction that is still novel must clear

- molecules ≥ 3 and reads ≥ 5
not read count alone — reads are PCR copies
- ≥ 0.1% of the reads at its own donor AND its own acceptor
- GT-AG always; GC-AG and AT-AC only with short-read confirmation in ≥ 2 replicates, or ≥ 50 reads
- no direct repeat ≥ 8 bp flanking the intron
RT template switching
- local FDR ≤ 0.05 from the calibrated model

the calibrated model

Ridge logistic → two-component mixture → local FDR, with Storey π_0 , and a monotone fit, stratified by whether the junction reuses an annotated splice site. Annotated junctions are the positive anchor; junctions on rejected reads are the negative one.

Fitted PER CONTIG — which is exactly what makes panel 0's parallelism safe: no calibration crosses a contig boundary, so splitting the run changes no threshold and no decision.

lfdr, posterior_real_in junctions.tsv

4 Chain identity

Reads are grouped on the exact curated intron chain, stored in transcript orientation so that a 5'-truncated read is always a SUFFIX of its parent. The representative is ALWAYS the maximal chain: merges only ever run from a shorter chain into a longer one.

chain matches an annotated chain

GENCODE
model

FSM — the model takes that transcript's GENCODE identity.
ENST00000229239..10

unannotated chain that is a SUFFIX of an annotated one

GENCODE
reads

5' truncation — merged into the parent whatever the support ratio says. The truncated form routinely outnumbered the intact one.

...unless its 5' end sits at an annotated TSS

GENCODE
model
annotated TSS

A real alternative start — kept as its own model.
ENST..alt5end1

independent chain, every junction annotated

GENCODE
model

NIC — a novel combination of known splice sites.
GAPDH|NIC_1

independent chain carrying a novel junction

GENCODE
model
novel junction

NNC — reaches this point only if that junction survived panel 3.
GAPDH|NNC_1

an intron is retained, with exonic flank on BOTH sides

GENCODE
model

IR. Flank on both sides is what separates a retained intron from a plain 3' truncation that happens to stop inside one.
ENST..|IR1

the chain spans two annotated genes

GENCODE
model
gene A · gene B

READTHROUGH — its own category, not a novel isoform of either gene.
GENEA|readthrough1

neither side annotated and support is comparable

Genuinely ambiguous — kept separate and flagged, never guessed.
flagged_in_group_chains.tsv

5 Terminal ends

3' is the anchored end under oligo-dT, so it is peak-clustered on MOLECULE mass. 5' is the degraded end: a model's 5' end is the 90th percentile of its members, never the minimum. Ends are compared against EVERY transcript sharing the chain, not against one designated parent.

peak matches the parent transcript's annotated 3' end

GENCODE
model

FSM.
ENST00000229239..10

peak matches ANOTHER transcript of the same gene

GENCODE
model
3' end of tx B

An annotated alternative end, not a discovery. Checking only the parent undercuts these: 35% of 3' ends match it, 72% match some transcript of the gene.
ENST..alt3end1

novel peak, ≥ 5% of the group, polyA tail on its molecules

GENCODE
model
AAAA..tail

A candidate alternative polyA site. In 0.5.x this is no longer the last word — every model that lands here is re-examined in panel 9.
ENST..|novel3end1

peak rejected, but an annotated end is within 300 bp

GENCODE
model
≤ 300 bp

Snapped. Annotation REFINING an observed coordinate, never manufacturing a distant one: the median model moves 5 bp and none moves more than 300.
FSM

A-rich genomic window downstream, no tail on the reads

model
A-rich, no tail

Discarded as internal priming. The oligo-dT primed on genomic A's, which ALIGN rather than clip — so an internally primed molecule shows a read pile and no tail.

5' end upstream of the annotated TSS

A credible alternative start — unless it crosses an annotated junction, which makes it a retained first intron instead.
ENST..ext5end1

parent transcript vs associated transcript

A PARENT is only over a GENCODE transcript whose intron chain equals the model's. NIC and NNC have a novel chain, so they have NO parent. associated transcript is merely the closest reference transcript and is annotation only — it never feeds back into calibration.

6 Mono-exon track

Runs entirely separately. Unspliced reads only ever group with each other, so a single-exon stub can never absorb spliced read mass.

inside the last exon or 3'UTR of a known gene

GENCODE
model

Trusted: with polyA capture this is RT drop-off in a long 3'UTR, and the 3' end is still correct.
ENST..|3UTRfrag1

inside a gene body, not terminal

GENCODE
model
AATAAA

Needs an actual polyA site — otherwise discarded.
GENE|APAL1

intergenic

Needs a polyA site and a higher support floor.
NOVELMONO_chr1_3

every mono-exon model is now re-judged

The track above asks "terminal exon or not". It never asks whether a single-exon transcript could exist in that gene at all. Panel 8 does.

7 SQANTI3 view

Since 0.4.2 every model also carries SQANTI3's own vocabulary, written BESIDE category rather than instead of it, so the two labellings are comparable on the same models.

- structural_category — FSM / ISM / NIC / NNC / fusion / genic / intergenic
- subcategory — SQANTI3's own strings
- terminal_class — where the 3' end sits, independent of the chain

The two vocabularies do not map one-to-one. end3_annotated, end3_novel, end5_alt and end5_extended all have an intron chain identical to an annotated transcript, so SQANTI3 calls them FSM with an alternative-end subcategory — not four kinds of novelty. Measured on the nine 0.5.1 runs: FSM is 17.6–20.2% of models under flightcollapse's labels and 54.7–58.9% under SQANTI3's structural_category, subcategory, terminal_class in models.tsv

8 Contig-parallel execution

The first pass is unchanged: same reads, same junction curation, same suffix resolution, same 3'-peak clustering, same models. The second pass runs afterwards on the catalogue the first pass produced, and may only keep a model, remove it, or replace it with an explicitly inferred one. It never builds a structure from scratch.

A filter folded into model construction cannot be turned off, cannot be compared against its own absence, and cannot report what it cost. This one writes one row per decision and can be disabled with a flag — - no -second-pass reproduces the 0.4.6 catalogue from the same binary.

-prefix>.secondpass.tsv — one row per decision, with its reason

BD144_kin, summed contig work	
wall clock at -j 8	0.65 h
speed-up	3.5x
floor set by the slowest contig (chr1)	997 s = 43% of wall

Longest-first dispatch, so the tail is not a whole chr1 long. Memory, not cores, is the binding constraint: peak RSS = workers × per-contig peak, and worker_count() clamps against MemAvailable.

0.5.1 · a killed worker no longer hangs the run

multiprocessing.Pool has no abrupt-death detection. When a worker is SIGKILLed — the OOM killer is how that happens here — the pool quietly forks a replacement, but the task it was holding produces no result and `Unap_order`ed waits for it forever.

Observed: BD144_kin sat at 23 of 24 contigs for two and a half hours with every worker idle and nothing in the log. The parent now tracks which pid owns which contig, notices when one disappears, and re-runs that contig itself — serially, in the parent, after the pool closes, because if the cause was memory then re-dispatching into a full pool would just kill it again.

9 Rule 1 — mono-exonic models judged against their gene

Five classes, decided from the annotation alone. "Spans an intron" uses constitutively intronic sequence — inside the gene span and exonic in iso form. Using any one transcript's introns would call an ordinary alternative-exon model a read-through, because one isoform's exon is another's intron all over the genome.

A · gene is mono-exonic only → KEEP

GENCODE
model

B · gene has mono + multi isoforms → KEEP

tx 1
tx 2
model

C · multi-exon gene, model spans an intron → REPLACE

GENCODE
model
≥ 10 bp of constitutively intronic sequence

D · multi-exon gene, model inside ONE exon → REPLACE

GENCODE
model

E · multi-exon gene, anything else → REPLACE

Wholly intronic, or running off the end of the gene.

X · no gene (intergenic) — out of scope for A–E

Left to the existing monoxon_intergenic handling. intergenic_policy = keep.

a replaced model does not simply vanish

removed in place
the gene's best-supported isoform, marked INFERRED

The gene's most-supported emitted multi-exon model is recorded in its place, falling back to the reference canonical (MANE_Select > Ensembl_canonical > basic > longest) when the gene has no emitted spliced model. The removed model's reads move onto it, so read accounting stays exact.

The category, a flag and an inferred=1 column all say the same thing three times, because the one failure this must not have is an inferred model being read as an observed one.

INFERRED · inferred · inferred_source · inferred_transcript

measured across the nine 0.5.1 runs

mono-exon models replaced (CID/E)	167,467
...from this sample's own data	156,902 (93.7%)
...from the reference canonical	10,565 (6.3%)
distinct INFERRED models created	7,715
reads folded onto inferred models, BD144_kin	675,114
catalogue shrinkage, whole second pass	-8.5% → -13.0%

BD144_kin classes: A 3,006 · B 2,283 · C 10,734 · D 4,959 · E 5,548 · X 4,816. Roughly four in five in-gene mono-exon models are CID/E.

Rule 2 — end3_novel must earn its 3' end

On SIRV Set 4 this category was wrong 12/12 in the reference arm and 11/11 in the masked arm. The mechanism is internal priming: oligo-dT anneals to a genomic A-stretch inside the transcript, and the resulting pile accumulates support exactly like a real cleavage site.

Three things that cannot separate them. A read floor cannot — false ends reach p90 548 reads while true ones go down to 5. tail_molecule_frac cannot — 0.96 vs 1.00 on BD176c — because it scores an A-run without asking what the genome says underneath. And cross-sample reproducibility cannot: an internal-priming site recurs in every library made from the same genome, so a model called in all three HEK samples is not thereby real.

six tiers, strongest first — the first that speaks decides	
1 · atlas — end sits on a catalogued polyA site	63.5%
2 · nontemplated — reads' tails exceed the genome's A-run	35.2%
3 · composition — hexamer + UGU-rich DSE + tight peak	0.1%
4 · short_read — coverage demonstrably drops across the end	0.0%
5 · calibrated — terminal local FDR clears a ceiling	off by default
6 · gene_dominant — sole or best-supported model of its gene	1.2%

Share of the 219,368 end3_novel models kept across the nine runs. The tier is written on every judged model, so a downstream filter can pick its own operating point without re-running anything.
end3_second_pass_tier in models.tsv

tier 2 · the templated-tail test — the only per-MOLECULE discriminator

genome
real
read A-run runs far past the genomic A-run → non-templated excess ≥ 8 bp

genome
primed
the aligner consumed the genomic A's — no excess, no tail

A genuine poly(A) tail is not in the genome. Everything needed was already being measured — reads, tail_len per read since 0.1.17, genome_fetch since the beginning — and nothing compared the two.

Defaults: excess ≥ 8 bp on ≥ 50% of the model's reads, scanning 60 bp downstream. This is why the collapse must read the UNTRIMMED dedup BAM: a trimmed read has no terminal A-run and the tier would silently have no opinion.

genomic_a_run_3p · median_tail_excess · frac_reads_nontemplated

tier 3 · a contrast, not an absolute cut

A real 3' end has a UGU-rich downstream element at +5...+30; an internal-priming site is A-rich. The tier scores (G+U) — A ≥ 0.10 rather than an absolute G+U threshold, because the window is ~25 bases and an absolute 0.45 cut is one base away from 0.40 — neutral sequence crosses it by chance about half the time. Neutral sequence sits near +0.25; an internal-priming site goes sharply negative.

A single per_c_downstream number cannot express this: on BD176c it moves only 20 → 25 between FSM and end3_novel. Peak dispersion does separate them — FSM 4.1 bp against end3_novel 34.6 bp — so a tight peak is required alongside.

why gene_dominant is LAST

Used first it selects against the biology. A genuine proximal polyA site usually carries a minority of a gene's molecules, while a strong internal-priming site in a well-expressed gene is exactly the thing that becomes dominant. It exists to rescue models the evidence tiers cannot speak to — 1.2% of survivors — not to adjudicate the ones they can.

what the gate actually costs

end3_novel models judged, nine runs	239,695
kept	219,368
dropped	7.3% — 10.0% per sample
reads folded onto a surviving sibling, BD144_kin	103,393
reads left unassigned by the second pass	0

A dropped model's reads move to the best compatible survivor — same intron chain first, then the gene's best model — so the read accounting closes.

the separation is the point

A contig's result depends on no other contig, and all three calibrated models are fitted per contig already — so distributing contigs across processes changes no calibration, no threshold and no decision.

What is genuinely shared is small and all of it lives in the parent: model IDs and the PB N gene numbering (assigned in contig order), read_stat and group (stitched from per-contig shards), and the molecule matrix (accumulated from contig shards). For k only — under spawn every worker would re-read the GTF, which costs more than the parallelism saves.

BD144_kin, summed contig work
wall clock at -j 8 0.65 h
speed-up 3.5x
floor set by the slowest contig (chr1) 997 s = 43% of wall

11 Rule 1 — same chain, 3' ends within 200 bp

Models are partitioned by contig, strand and exact intron chain, so the rule can never merge two splice structures. Within a partition the representative is the annotated 3' end if there is one, then the best supported, then the longest. Everything within the tolerance of that representative is absorbed.

new default 200 bp in 0.6.3 · 500 bp in 0.6.0–0.6.2 · 0 merges only identical ends · -1 switches the rule off

identical chain, 3' ends 150 bp apart → MERGE

GENCODE
model A
model B

B's 3' end is 150 bp past A's · A carries the annotated end

B's reads move onto A and B's molecules are recounted from them. Nothing is discarded; the catalogue is one model shorter.
posthoc_end3_absorbed = 1

5'-truncated copy → FOLDED

model A
model B
same chain, same 3' end, different TSS — these now merge

The classic 5' truncation: a shorter chain that is a suffix of a longer one, with nothing of its own at either end.
posthoc_fragments_absorbed

alternative first exon inside an intron → KEPT

container
model
its first exon starts in the container's intron — outside the ≤50 bp slack

This is what the terminal-exon slack is for. A chain that is a sub-chain but whose own first exon starts somewhere the container has no exon is an alternative start, not a truncation.
kept as its own model

mono-exon inside one exon → FOLDED

container
model
wholly inside one exon of a 2+-better model

A single-exon model has no chain, so containment is judged on the exon alone.

exact annotated chain (FSM) → PROTECTED

GENCODE
model
a transcript the annotation asserts exists

FSM is never removed, whatever contains it. A reference-free rule deleting a reference transcript would be the filter overreaching into territory it has no evidence about.
protected_categories = [FSM]

container only 1.6x better → BOTH KEPT

container
model
50 molecules against 32 — below the 2x bar

Comparable support means the shorter form is not obviously debris. The ratio is the whole of the evidence the rule has, so it is held to deliberately.
subchain_ratio = 2.0

a container that is itself removed

Decisions are taken in one pass on pre-filter supports, then resolved: a fragment whose container was itself folded away lands on the final survivor, with cycle protection. Reads never stop at a model that is not in the catalogue.
into_model_id in {prefix}.posthoc.tsv

12 Rule 2 — reference-free sub-chain (fragment) filter

A model is removed, and its reads folded into the container, when its ordered intron chain is a contiguous sub-chain of a longer model on the same contig and strand (junctions ≤5 bp), its terminal exons lie inside the matching exons of that model (≤50 bp), and the container carries ≥ 2x its support.

no annotation is consulted at any point · decisions are taken in one pass on pre-filter supports

Why the two ends are not treated alike

The 3' end is a position — the mode of a cleavage-site peak — and only a tolerance can say whether two of them are the same site. The 5' end is a containment relation between two intron chains, which the sub-chain filter decides from structure alone, with no annotation and no distance threshold.

this asymmetry is the design, not an accident of tuning

13 Why the two ends are not treated alike

The 3' end is a position — the mode of a cleavage-site peak — and only a tolerance can say whether two of them are the same site. The 5' end is a containment relation between two intron chains, which the sub-chain filter decides from structure alone, with no annotation and no distance threshold.

this asymmetry is the design, not an accident of tuning

what the guard being off costs, and buys

model A
model B
same chain, same 3' end, different TSS — these now merge

With end5_window on, that pair never merged. But neither did a 5'-truncated copy, because it shares the chain exactly and so is invisible to the sub-chain filter too. 0.6.3 accepts the first case to fix the second: across the three libraries it freed 24–5.664 merges per arm for at most 0.03 pp of mass.
posthoc_end5_window = -1 · =100 restores the guard

200 bp against 500 bp, on three libraries

models	499,553 → 418,097	→ 499,553 → 392,017
	-16.3%	-21.5%
reproducible read mass	-0.32 pp	-1.11 pp
at ≤50 reads, ≥100 bc	+0.11 pp	-0.25 pp
cells where mass rises	34 of 81	17 of 81
splice chains	122,270	122,174

an outside check nobody designed for

ISM, 3' fragment	6,542 → 2,649	-60%
ISM, 5' fragment	20,590 → 11,616	-44%
FSM, alt 3' end	44,985 → 37,469	-17%
FSM (reference mass)	33,018 → 31,581	-4%

pigeon / SQANTI3 classes, BD144_kin

The sub-chain filter never consults an annotation, and pigeon is nothing but an annotation — yet the classes it empties are exactly the fragment classes, while the 3' tolerance's signature appears in FSM with an alternative 3' end. A rule cutting real transcripts would not sort itself along a reference it cannot see.
reference-free rule, reference-based verdict

where the stage sits

Step 8b: last on every contig, after the second pass and before read bookkeeping — so group.txt, read_stat, the abundance tables and the cell × isoform matrix all describe the catalogue that is actually written. The two rules run 3' first, so the ratio in rule 2 is tested on consolidated counts.

-no -posthoc reproduces 0.5.1 byte for byte

CONSOLIDATION — new in 0.6.0, retuned in 0.6.3

runs last on every contig · both rules only MOVE support between existing models, so the read totals are identical with the stage on and off

14 Rule 1 — same chain, 3' ends within 200 bp

Models are partitioned by contig, strand and exact intron chain, so the rule can never merge two splice structures. Within a partition the representative is the annotated 3' end if there is one, then the best supported, then the longest. Everything within the tolerance of that representative is absorbed.

new default 200 bp in 0.6.3 · 500 bp in 0.6.0–0.6.2 · 0 merges only identical ends · -1 switches the rule off

identical chain, 3' ends 400 bp apart → KEPT

model A
model B
beyond the tolerance — two 3' ends, two models

At 500 bp these would have merged. That is the whole of the difference between the two defaults: 26,090 further merges across the three HEK libraries, bought with 0.79 pp of reproducible read mass.

anchored, not chained

A rep B C
A..B 180 bp · B..C 180 bp · A..C 360 bp

B is absorbed because it is within 200 bp of A. C is not, even though it is within 200 bp of B. Single-linkage chaining would let one cluster span the whole locus; anchoring to the representative bounds every cluster at the tolerance.
C starts the next cluster

one junction differs → NEVER COMPARED

model A
model B
one extra junction — a different partition

The partition key is the exact intron chain, so no tolerance, however large, can merge two splice structures.

INFERRED models are untouched

A model the second pass substituted for a rejected one is neither absorbed nor used as a representative. It is not evidence about a 3' end — it is a stand-in for a model that failed.
category = INFERRED

Outputs

sorted GTF / GFF · read_stat.txt · group.txt · abundance.txt · file_count.txt · models.tsv · junctions.tsv (with drop_reason) · group_chains.tsv · secondpass.tsv · monoxon_rejected.tsv · cell × transcript UMI matrix · qc.json / qc.md

flightcollapse v0.6.3 · snapshot